Examiner only

Answer **all** questions.

1. (a) Background radiation occurs due to the decay of an unstable nucleus releasing alpha, beta or gamma radiation.

Complete the table below to state what each type of radiation is. [3]

Type of radiation	Symbol	What it is
Alpha	${}^{4}_{2}\alpha$	....
Beta	${}^{0}_{-1}\beta$	....
Gamma	γ	....

(b) A teacher demonstrates how to determine the background radiation count in her laboratory.
She uses a radiation detector and measures 18 counts in 30 seconds.

(i) Determine the background radiation count in counts per second. [1]

Background radiation = ... counts per second

(ii) Suggest **two** ways that the teacher could improve her measurement of background radiation. [2]

1. ....

2. ....

(iii) Explain why the background radiation count is much higher in some parts of the country than others. [2]

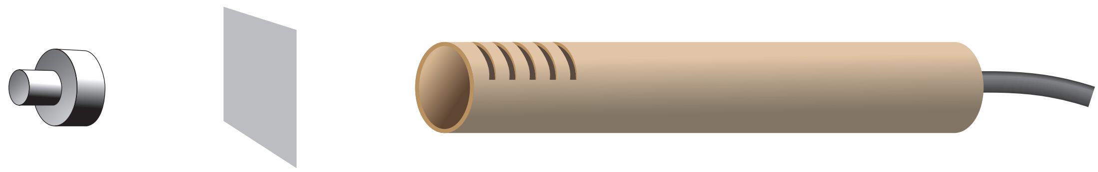
....

....

....

Examiner only

(c) The teacher now demonstrates an experiment to identify the radiation emitted by different sources.
The count rate is measured, first with no absorber present and then with paper and aluminium absorbers separately.



The results are shown in the table below. They are corrected for background radiation.

Source	Count rate (counts per second)		
	No absorber	Paper	Thin aluminium
1	312	313	312
2	389	57	0

(i) Write **yes (Y) or no (N)** in each box to show which type(s) of radiation is (are) emitted by each source. [4]

	Alpha	Beta	Gamma
Source 1			
Source 2			

(ii) Explain **one** change the teacher could make to extend the investigation to confirm to the class the conclusions made. [2]

....

....

....

14

3420UB01
05